AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 4 · STUDENT LESSON

Activation functions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Compute a score. Apply a function. State the decision rule.

Trace fixed activation functions and explain their information and evidence limits.

Keep one habit: Complete table → labelled curves → pictured ReLU/identity chain → stable-code probes → independent new chain.

MAIN LESSON · SECTION 1

What happens to a score before the next layer?

> Compare fixed functions on the same values before changing a whole network.

Learning goals

I can trace identity **ReLU** and **sigmoid** on a numerical score.

I can explain how a nonlinear activation changes a computation.

I can distinguish an activation **output** from a final class or verified probability.

Recall the affine score

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- For z = 2 times width - height - 3 at [3, 3], what are the score and class under z >= 0?

My starting explanation

End goal

By the end, I can explain **activation function**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Trace identity **ReLU** and **sigmoid** on a numerical score

Explain how a nonlinear activation changes a computation

Distinguish an activation **output** from a final class or verified probability

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

A score goes into a function;an activation comes out

> The output may feed another layer rather than directly name a class.

Declare the input and compare identity/ReLU

First compute an affine score z = w dot x + b. An activation then returns a = f(z). In a network this value may become an **input** to another layer; it is not automatically a class **prediction**.

We will compare three functions using the same incoming scores so that only the activation changes.

Identity returns z unchanged. **ReLU**, pronounced ree-loo, returns the larger of zero and z. Thus **ReLU**(-2) = 0, **ReLU**(0) = 0 and **ReLU**(2) = 2. Positive scores keep their magnitude; **ReLU** does not turn every positive value into 1.

Complete same-score comparison

Incoming z	Identity output	ReLU output
-2	-2	0
0	0	0
2	2	2

Distinguish fixed functions from training

These are fixed arithmetic functions. Selecting a function or changing its **input** is not itself a **training** procedure. Later weeks will introduce losses and parameter updates.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Read the function,the value and the class rule separately

> A range between zero and one is a mathematical property,not a reliability result.

Sigmoid definition and finite-input interpretation

Sigmoid is 1 / (1 + exp(-z)). Here exp(t) means e raised to power t, where e is approximately 2.718. A calculator or the supplied Python code can evaluate it; no calculus is required this week.

For z = -2, 0 and 2, **sigmoid output**s are approximately 0.1192, 0.5 and 0.8808.

For every finite mathematical **input**, **sigmoid** is strictly between 0 and 1, though computer rounding can produce an endpoint for extreme scores.

Negative **input**s produce values below 0.5, zero gives 0.5, and positive **input**s give values above 0.5. A class rule still needs to be stated: if class 1 means **sigmoid**(z) >= 0.5, these three **prediction**s are 0, 1 and 1.

Complete three-function table

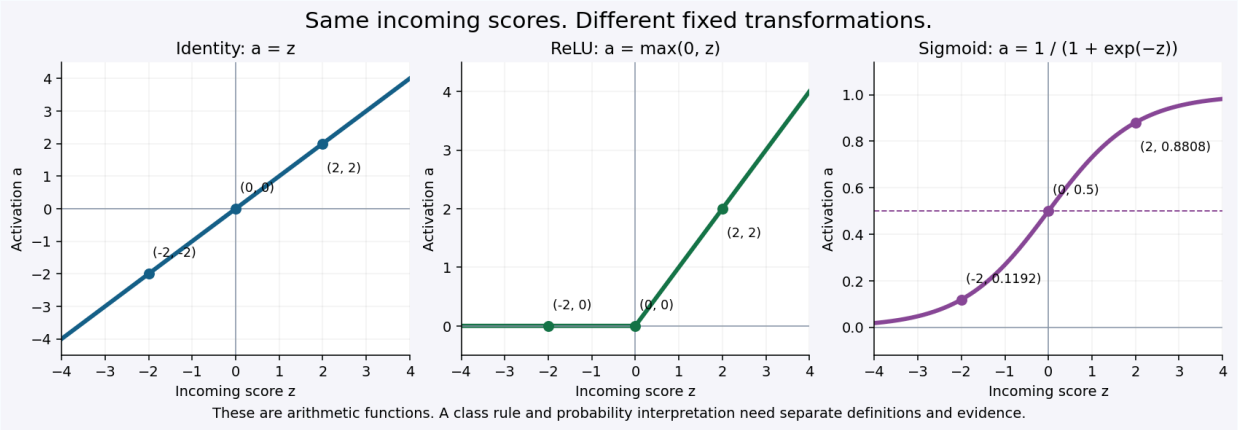
Incoming score z · Identity z · ReLU max(0, z) · Sigmoid ≈ · Class if sigmoid≥0.5

> **Incoming score z:** -2
>
> **Identity z:** -2
>
> **ReLU max(0, z):** 0
>
> **Sigmoid ≈:** 0.1192
>
> **Class if sigmoid≥0.5:** 0

> **Incoming score z:** 0
>
> **Identity z:** 0
>
> **ReLU max(0, z):** 0
>
> **Sigmoid ≈:** 0.5
>
> **Class if sigmoid≥0.5:** 1

> **Incoming score z:** 2
>
> **Identity z:** 2
>
> **ReLU max(0, z):** 2
>
> **Sigmoid ≈:** 0.8808
>
> **Class if sigmoid≥0.5:** 1

Labelled function graphs



Stable runnable calculation and evidence limits

```
from math import exp

def sigmoid(z):
    if z >= 0:
        return 1 / (1 + exp(-z))
    e = exp(z)
    return e / (1 + e)

for z in [-2, 0, 2]:
    print(z, max(0, z), round(sigmoid(z), 4))
```

The two algebraically equivalent branches avoid taking exp of a large positive number. A value within [0, 1] is not automatically a reliable probability: its interpretation depends on the fitted task and evaluation **evidence**.

Connect a graph point to a function evaluation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Change only the activation while keeping the rest fixed

> Predict which intermediate values survive.

Current class-rule task and response

- Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured identity replacement

- Change: Replace **ReLU** with identity.

▼ Check after predicting

Prediction: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace what the next step actually receives

> Once different inputs map to the same h,the final fixed step cannot recover their difference.

Complete picture formulas,inputs and scope

See what **ReLU** removes before the next weighted step.

Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses z=x−3, h=ReLU(z), y=3h+2. Inputs1/3/4 give z−2/0/1, h0/0/1, y2/2/5. Inputs1 and3 have become indistinguishable to this final step because both pass h0. Replace only ReLU with identity:the same z values survive as h−2/0/1, y−4/2/5. This is not the independent chain, which supplies different constants and inputs. No fitting or observed real-world success is represented.

Trace preactivation,activation and final value

AI ACADEMY / AI-5/4 · 02 · TRACE THE CASE

Follow the deciding steps



01 Compute preactivation

z values are -2, 0, 1.



02 Apply ReLU

h values are 0, 0, 1; the negative value is replaced by0.



03 Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute preactivation	z values are -2, 0, 1.
2. Apply ReLU	h values are 0, 0, 1; the negative value is replaced by0.
3. Compute output	y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

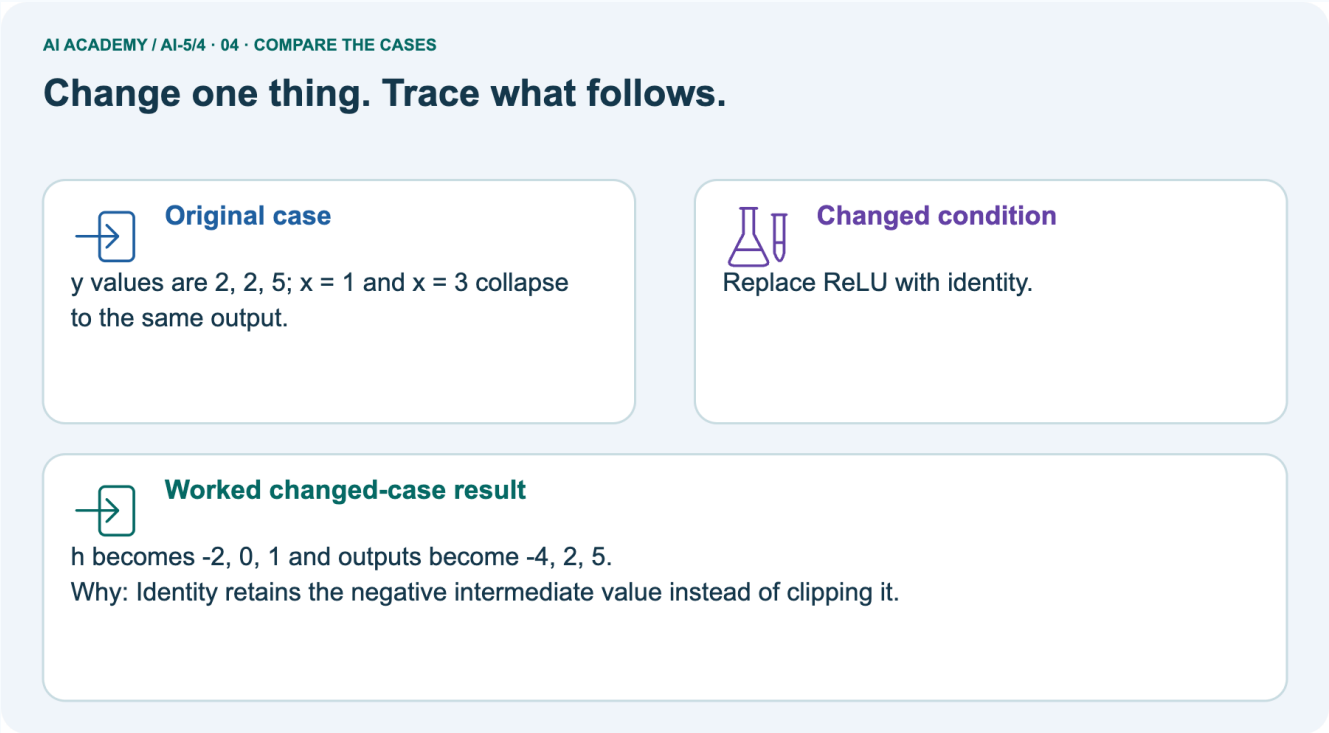
ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Predict identity in place of ReLU

- Change one thing: Replace **ReLU** with identity. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain information loss and training limits

- **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

- Read the labelled picture: Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Try the supported–1/+1functions

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
z is -2	Identity of z	-2
z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

For z = -1 and z = 1, give identity and ReLU outputs. Using sigmoid(-1) = 0.2689 and sigmoid(1) = 0.7311, give classes under the rule a >= 0.5.

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Activation functions

The facts we will use

AI ACADEMY / AI-5/4

ReLU hidden computation at x=1

Input 1 x weight 1 = 1

Add the products and bias -3: total = -2

ReLU: use max(0, total) -> output 0

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/4

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.





2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by0.





3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Show why a nonlinear step changes the possible function

> Use a concrete midpoint check instead of only a definition.

Complete midpoint and composed-chain comparisons

An affine function has one fixed slope. **ReLU** has slope zero on the negative side and slope one on the positive side, so no single straight line fits the whole function.

For a concrete check, **input**s -2, 0, 2 give **output**s 0, 0, 2. The average of the endpoint **output**s is 1, but the **output** at the midpoint 0 is 0. This differs from an affine function, which preserves midpoints.

For a one-**input** chain with identity between stages, h = 2x + 1 followed by y = 3h - 2 collapses to y = 6x + 1. Insert **ReLU** instead: h = max(0, 2x + 1), y = 3h - 2.

At x = -1, the identity chain gives -5 while the **ReLU** chain gives -2. At x = 1, both give 7. The nonlinear step can change the overall family of functions.

Try the midpoint repair

- Compare **ReLU** at **input**s -2, 0, 2. Give the average of the two endpoint **output**s and the **output** at the midpoint; explain what this shows.

> **State one repair and the checks you would rerun.**

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

Practise values,classes and explanation with less help

> Retain each intermediate value and the deciding rule.

Current task and response

- For scores -2, 0, 2, list identity and **ReLU output**s. Explain why **ReLU**(2) is not 1.

> **For z = -1 and z = 1, give identity and ReLU outputs. Using sigmoid(-1) = 0.2689 and sigmoid(1) = 0.7311, give classes under the rule a >= 0.5.**

> **Explain your prediction to a partner before comparing results.**

Supported W2record

Use workbook W2 for the connected example "**Activation function**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual misconception check

- Use the worked case for **Activation function**s. Proposed explanation: "**ReLU** returns a class label". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured information-loss reasoning

- Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Reserve the independent new chain

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Apply a declared class rule

Check your reasoning

Explain the picture

Investigate sign,zero,saturation and a declared class rule

> All supplied probes keep arithmetic and evidence status separate.

Complete current task and runnable sigmoid

Use the lesson **sigmoid** values for -2, 0, 2. Predict classes under a >= 0.5, then explain the zero-score case.

```
from math import exp

def sigmoid(z):
    if z >= 0:
        return 1 / (1 + exp(-z))
    e = exp(z)
    return e / (1 + e)

for z in [-2, 0, 2]:
    print(z, max(0, z), round(sigmoid(z), 4))
```

Normal,boundary,extreme,missing and rule-change probes

Complete activation probe	Expected output / action
Negative z−2	Identity−2, ReLU0, sigmoid≈.1192, class 0 under≥.5
Boundary z0	Identity0, ReLU0, sigmoid.5, class 1 under≥.5
Positive z2	Identity2, ReLU2, sigmoid≈.8808, class 1
Large finite z−1000 and1000	Stable code returns rounded sigmoid 0 and1; mathematical finite-input sigmoid remains strictly inside(0, 1)
Missing zNone	Outside finite-number input contract; request/validate before evaluating
Change class comparison≥.5to>.5at z0	Activation remains.5; class changes 1→0

Complete activation probe	Expected output / action
ReLU midpoint test−2/0/2	Endpoint outputs 0 and2 average1; midpoint output 0, so not one affine line

Predict,run or trace,and interpret

1. Predict each supplied probe, then run or trace the stable sigmoid code and max(0, z). Save actual values/errors separately from expectations. The branch for negative z uses exp(z), avoiding exponential overflow from−z when z is very negative. The function assumes finite numbers; it does not parse strings or validate missing/NaN inputs.

   Then trace the pictured chain and its identity replacement without changing any other constants. A new activation is a new computation; benefit would need a declared training/evaluation comparison.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: trace a new activation chain

> Use its own constants and inputs,then replace only the stated function.

Complete new independent chain

Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Trace a new chain independently

Attempt this without hints

Let z = x - 2, a = **ReLU**(z), y = 2a + 1. For x = 0, 2, 5, record every z, a and y. Then replace **ReLU** with identity and give all three y values. Which **input**s have the same **ReLU output**, and why?

> **My independent procedure, result and limitation**
>
> | |
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Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Record the activation and every intermediate value

> Use the actual project's formula and evidence.

Project artifact and evidence table

Add a labelled **activation function**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace identity **relu** and **sigmoid** on a numerical score.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Keep the actual run/trace and limits

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish by separating score,activation and decision

> Name one distinction the activation erased and one claim requiring evidence.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Trace identity **ReLU** and **sigmoid** on a numerical score

> **My deciding step, evidence and one remaining question**
>
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Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Activation function	A function applied to a computed score
ReLU	The function max(0, z), retaining positive scores and replacing negative scores with zero
Sigmoid	The function 1 / (1 + exp(-z)), mapping finite real scores into the interval between 0 and 1
Nonlinear function	A function that cannot be represented by one fixed slope and offset over its entire domain

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For scores -2, 0, 2, list identity and **ReLU output**s. Explain why **ReLU**(2) is not 1.
- Compare **ReLU** at **input**s -2, 0, 2. Give the average of the two endpoint **output**s and the **output** at the midpoint; explain what this shows.
- A model returns a **sigmoid** value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent z=x−2,ReLU,output chain for the first individual attempt.

Current explanation and response

- A nonlinear activation allows a network to represent relationships that a chain of affine operations alone cannot express. It does not guarantee that **training** will find a useful model or that more layers will help.

  ReLU can erase differences between negative scores; **sigmoid** can compress very large scores into nearly identical values. The architecture, data and **training** procedure still matter.

- Next week, we will compose layers while keeping every intermediate value and its role visible.

- A model returns a **sigmoid** value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

A learner reports sigmoid(2) is about 0.8808 and concludes the system will be correct 88 percent of the time for every future case. What does the calculation establish, and what evidence is still missing?

My limited claim and a useful next check

▼ Extra practice · Revisit the complete trace

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Three functions receive the same score

SAME INCOMING SCORE	APPLY ONE FUNCTION	ACTIVATION OUTPUT
z is -2	Identity of z	-2
z is -2	ReLU of z	0
z is -2	Sigmoid of z	About 0.1192

Each row starts again from the same score -2 and applies one function. The functions are alternatives here, not successive steps. These outputs are activation values, not class labels.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: See what **ReLU** removes before the next weighted step.

By the end, I can	How I will show it
Trace identity ReLU and sigmoid on a numerical score	Explain the deciding step, show a trace or test, and state one limit.
Explain how a nonlinear activation changes a computation	Explain the deciding step, show a trace or test, and state one limit.
Distinguish an activation output from a final class or verified probability	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- See what **ReLU** removes before the next weighted step.
- Which labelled step matches this detail? y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.
- 1. Apply **ReLU**
- 2. Compute **output**
- 3. Compute preactivation
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compute **output** y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. **ReLU** means keep a positive value and replace a negative one with zero.

The **output** layer receives that transformed value, not the original **input**. **Input**s one and three therefore both deliver zero to the final layer and produce two.

Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Changed-case question and optional hint

- Changed-case practice: Replace **ReLU** with identity. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. h becomes -2, 0, 1 and **output**s become -4, 2, 5.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: h becomes -2, 0, 1 and **output**s become -4, 2, 5. Identity retains the negative intermediate value instead of clipping it.

Repair and retain the record

Review Bias terms and decision boundaries (ai-5/3). Goal: Calculate a weighted score with a bias term. Short practice: Is B positive under the original or changed model?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Representing **input**s as vectors (ai-5/1). Goal: Encode a record using a fixed feature order and units. Short practice: Why does equal total brightness not make A and B identical **input**s?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

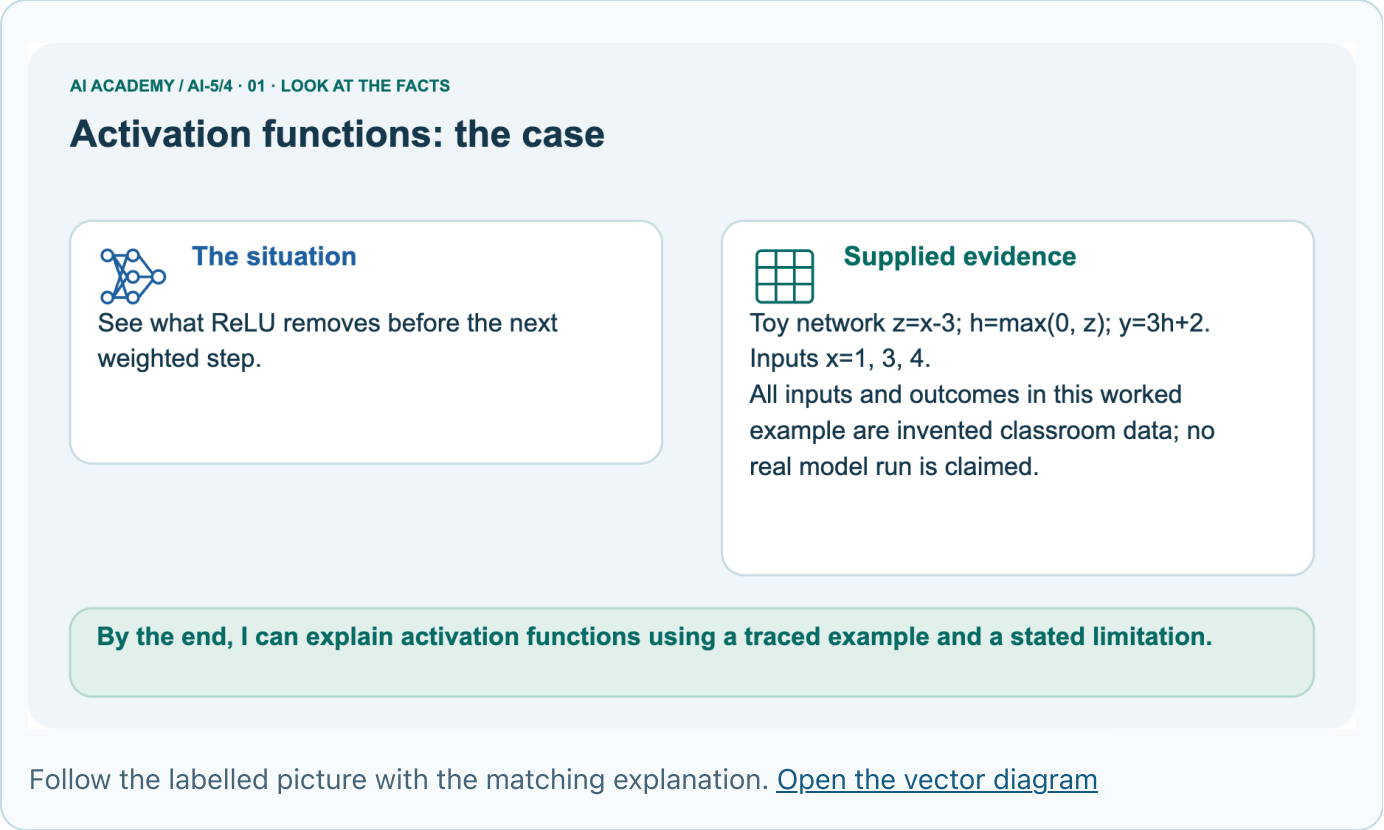
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Activation functions: the case



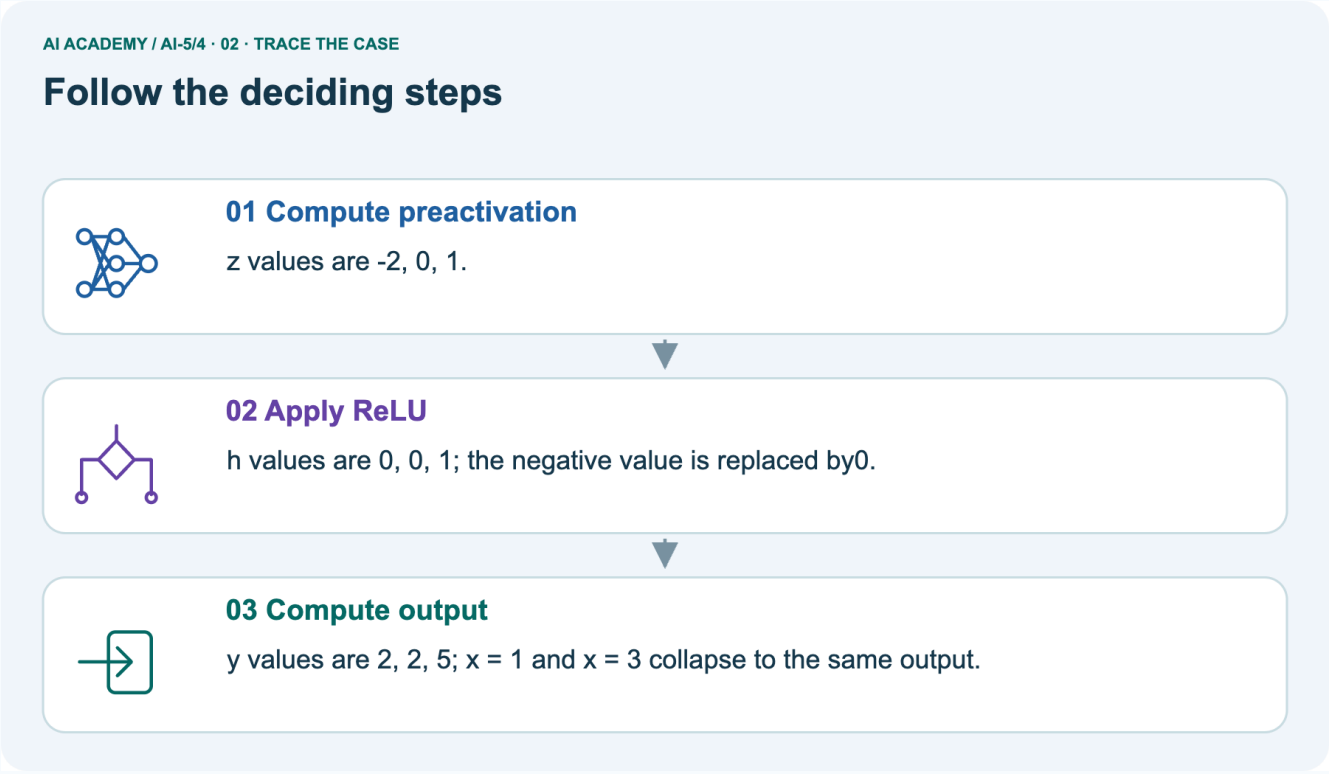
Follow the labelled picture with the matching explanation. Open the vector diagram

See what **ReLU** removes before the next weighted step. Supplied facts: Toy network z=x-3; h=max(0, z); y=3h+2. **Input**s x=1, 3, 4.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

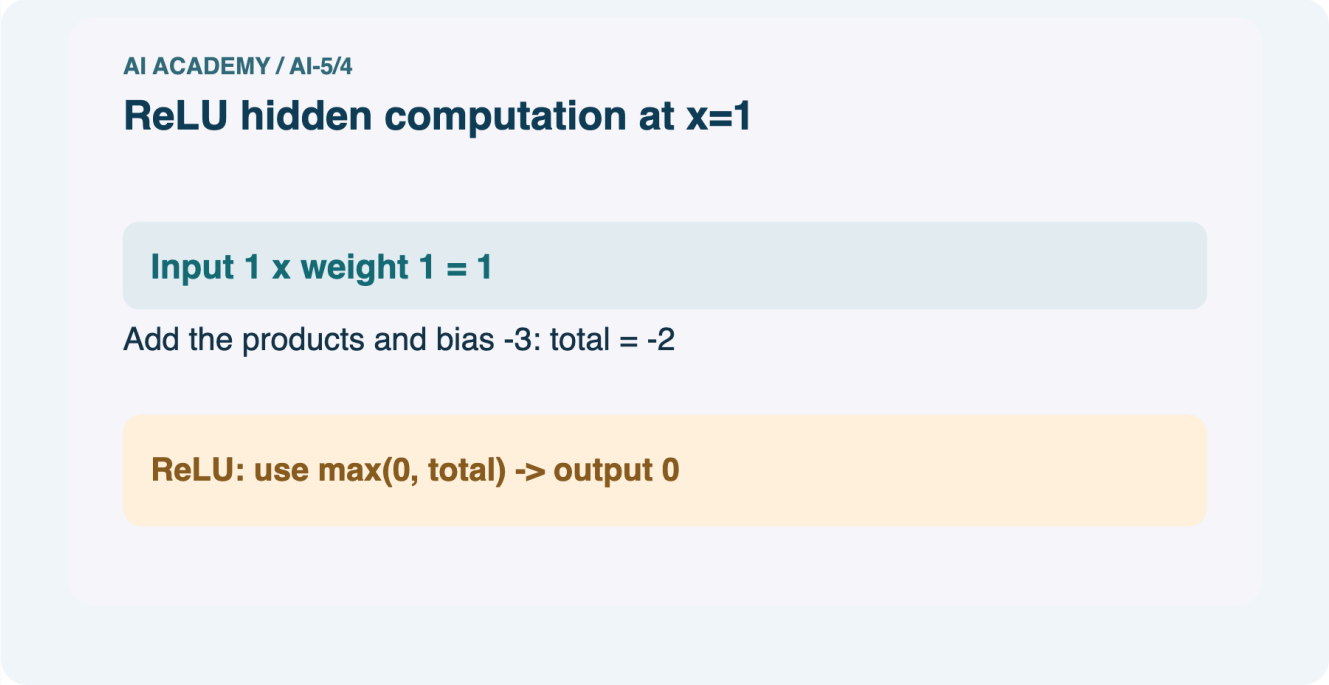


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute preactivation: z values are -2, 0, 1. → Apply **ReLU**: h values are 0, 0, 1; the negative value is replaced by0. → Compute **output**: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**.

Point to each numbered stage and explain which recorded **input** it uses.

03. ReLU hidden computation at x=1

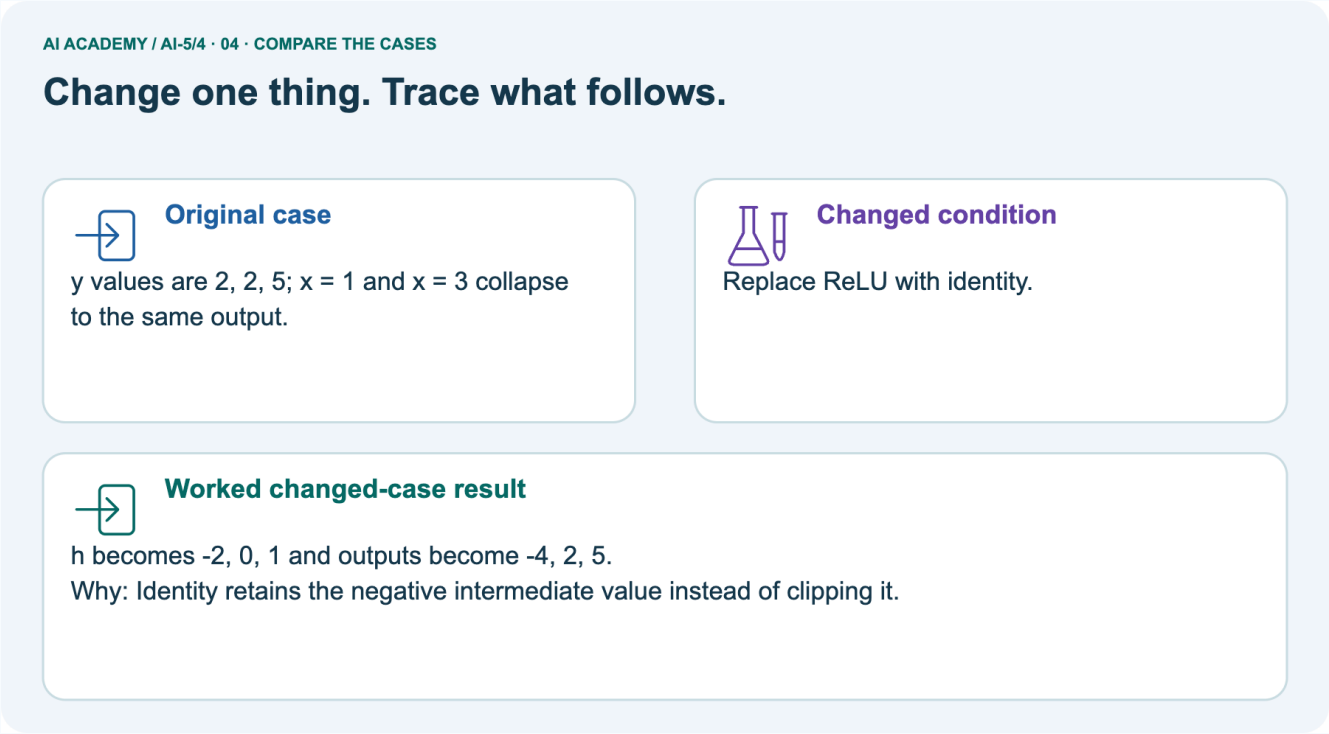


Follow the labelled picture with the matching explanation. Open the vector diagram

ReLU hidden computation at x=1. **Input**s: [1]; weights: [1]; bias: -3; activation: **relu**; **output**: 0.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



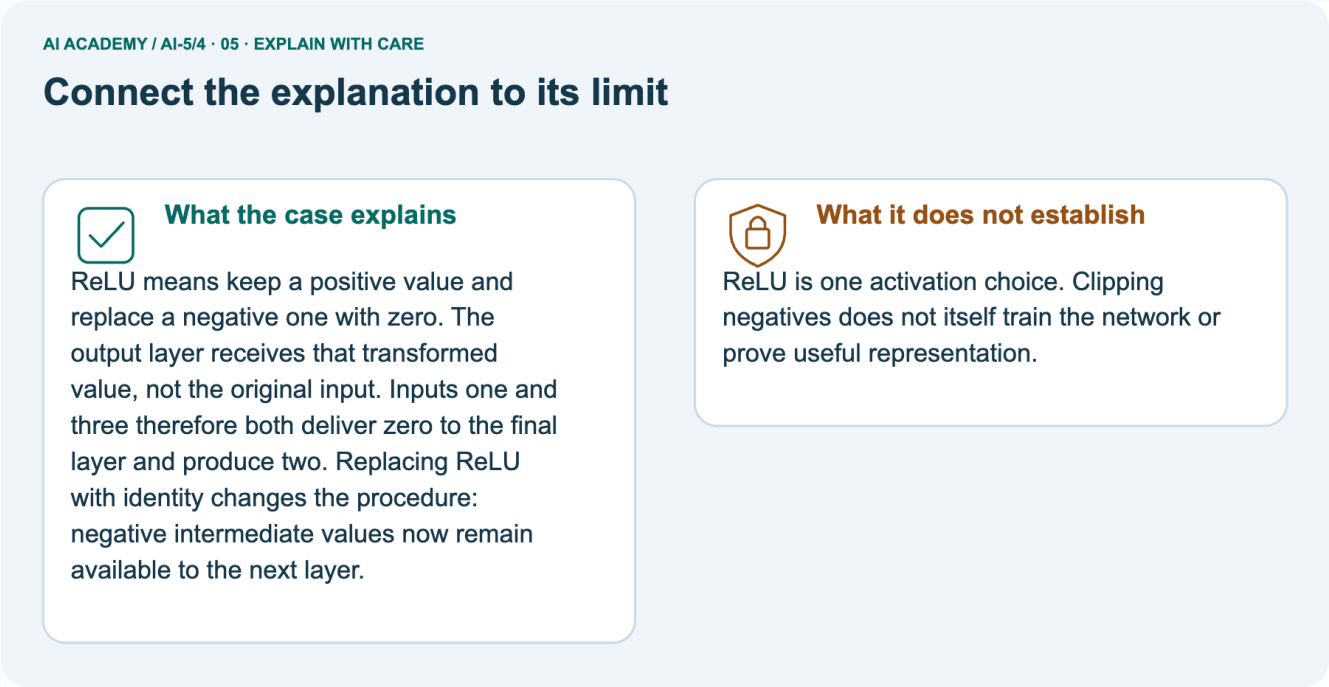
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y values are 2, 2, 5; x = 1 and x = 3 collapse to the same **output**. Changed condition: Replace **ReLU** with identity. Expected result: h becomes -2, 0, 1 and **output**s become -4, 2, 5.

Reason: Identity retains the negative intermediate value instead of clipping it.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

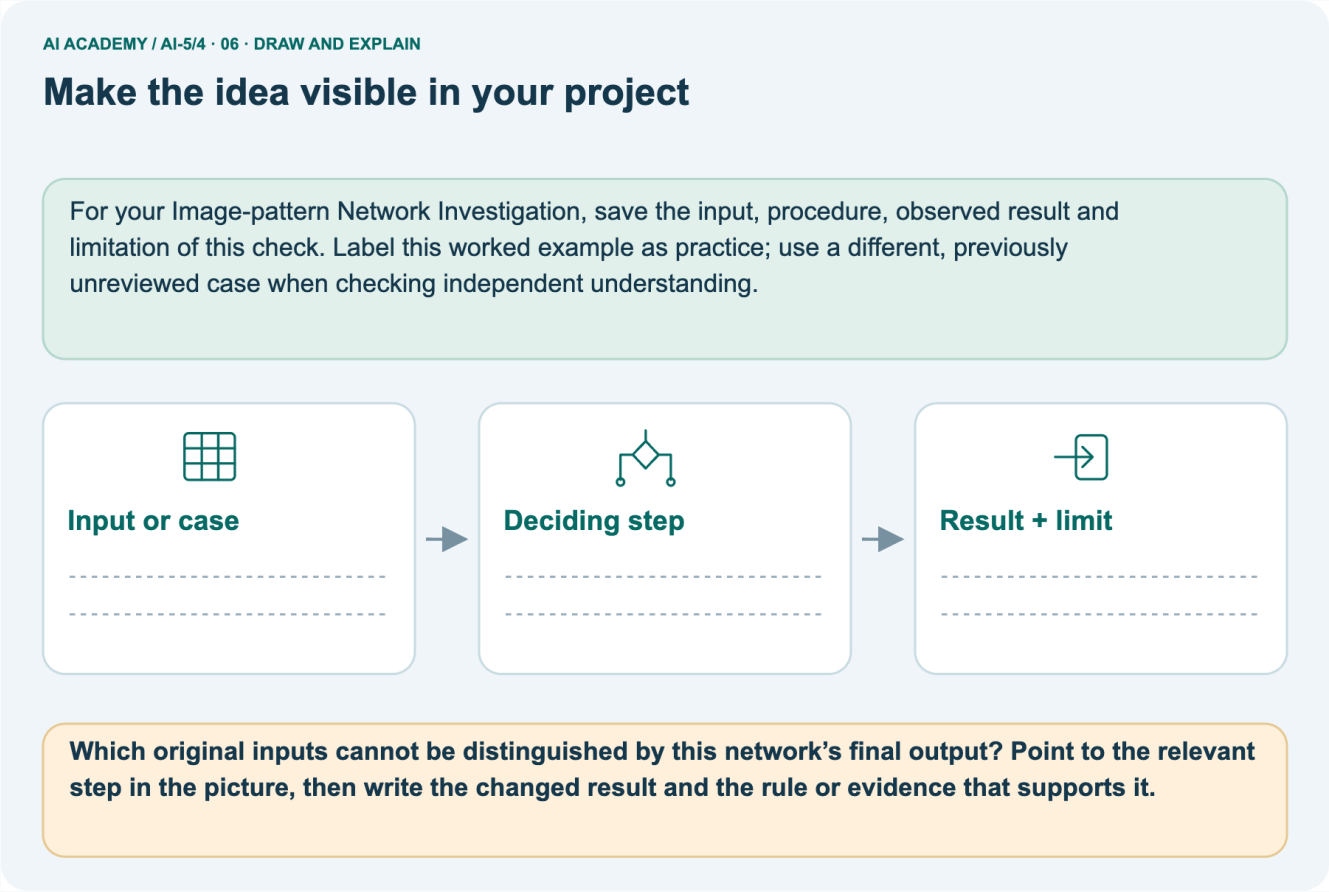
ReLU means keep a positive value and replace a negative one with zero. The **output** layer receives that transformed value, not the original **input**.

Inputs one and three therefore both deliver zero to the final layer and produce two. Replacing **ReLU** with identity changes the procedure: negative intermediate values now remain available to the next layer.

Limit: **ReLU** is one activation choice. Clipping negatives does not itself train the network or prove useful representation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which original **input**s cannot be distinguished by this network's final **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Activation functions: the case

02. Follow the deciding steps

03. ReLU hidden computation at x=1

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.